Lattice-Boltzmann Method for CFD

AMSC - HandsOn

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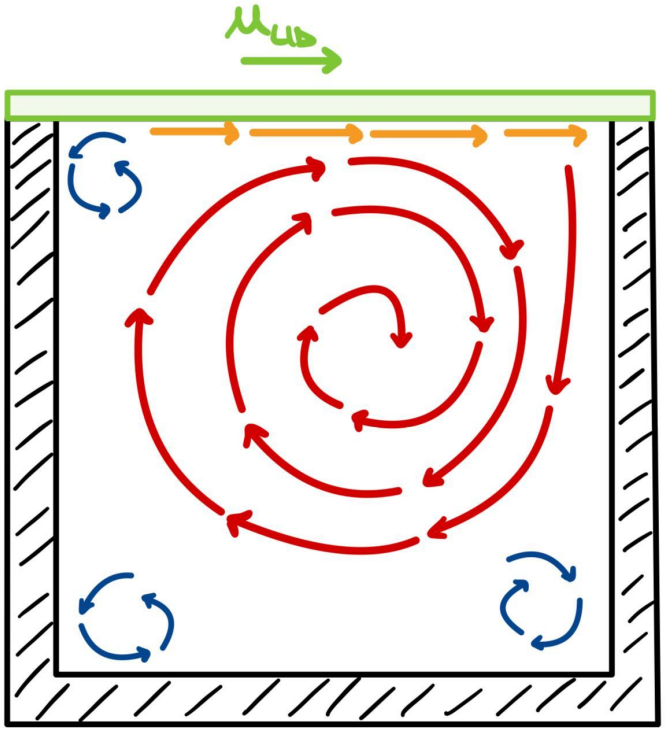
Simone Ferri, Giovanni Carpenedo, Cristiano Corona, Federico Pizzolato, Daniele Confalonieri

Lid-Driven Cavity Problem



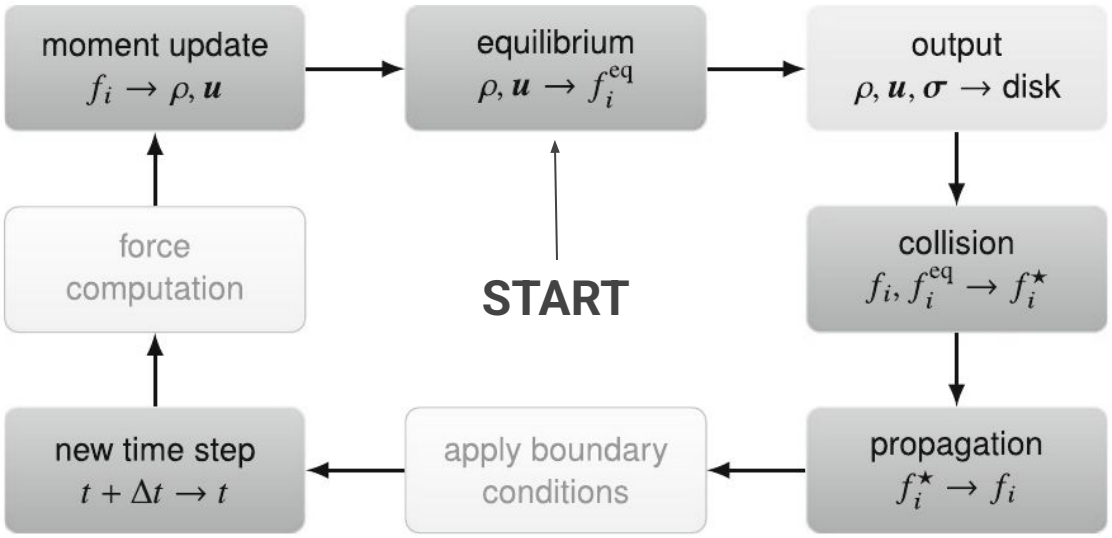
The Lid-Driven Cavity problem is a standard benchmark in CFD, in this simulation the fluid motion in a closed cavity is solely driven by the movement of one boundary wall called lid.

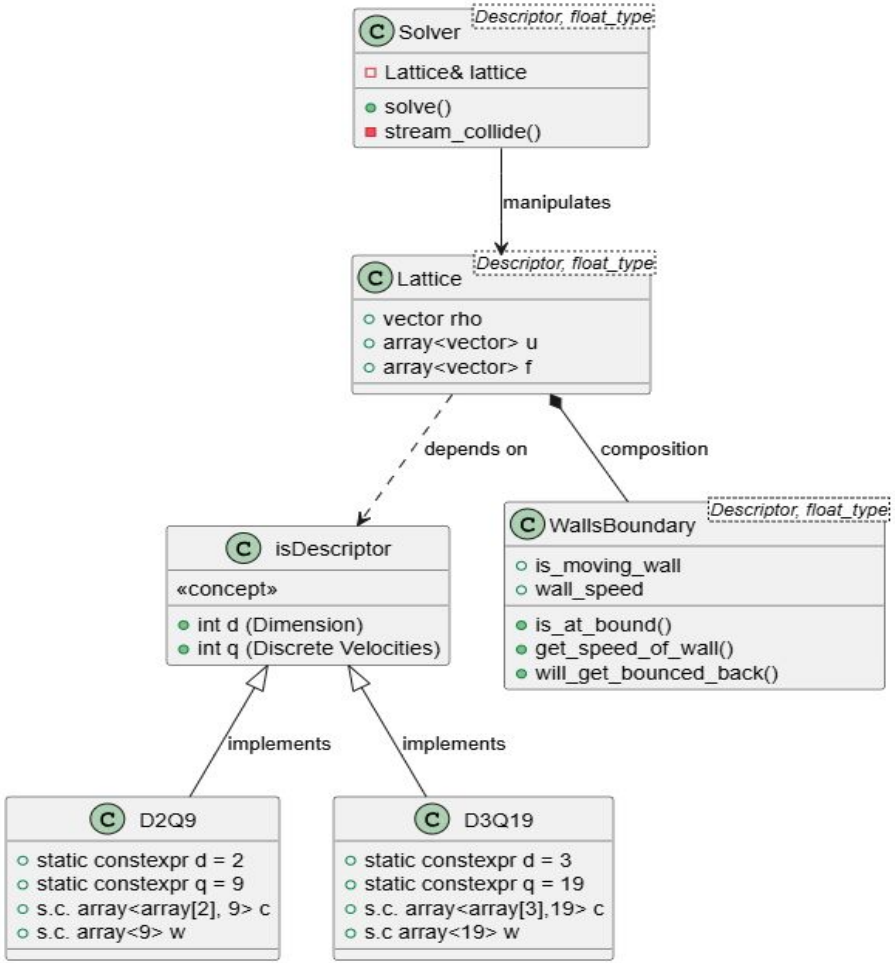
The moving lid drags the fluid along the wall driving the motion creating a vortex in the center and smaller ones near the corners as the Reynolds number increases.

The Reynolds number describes the relation between fluid viscosity and fluid velocity, at low Reynolds numbers the fluid flow is smooth while at high values it may be turbulent.

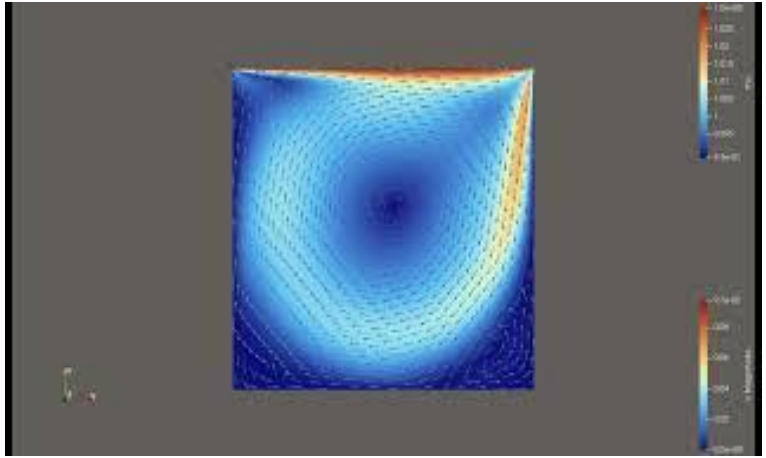
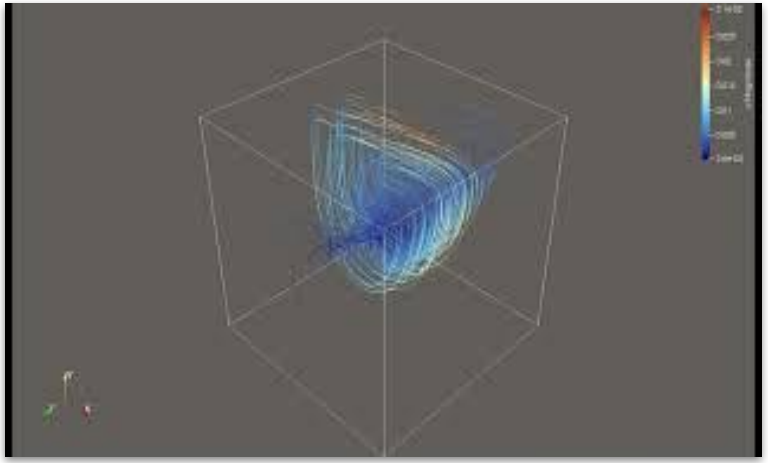
We want to simulate the behaviour of the fluid in a 2D squared box at Reynolds=100 using the Lattice-Boltzmann method due to its parallel nature and its ability to scale with the dimension of the grid.

LBM


moment update
$f_i \rightarrow \rho, \boldsymbol{u}$
equilibrium
$\rho, \boldsymbol{u} \rightarrow f_i^{\mathrm{eq}}$
output
$\rho, \boldsymbol{u}, \boldsymbol{\sigma} \rightarrow$ disk
force computation
START
collision
$f_i, f_i^{\mathrm{eq}} \rightarrow f_i^{\star}$
new time step
$t + \Delta t \rightarrow t$
apply boundary conditions
propagation
$f_i^{\star} \rightarrow f_i$



Project Structure

- Class templating to allow different floating point precisions and various descriptors to modify the lattice geometry.
- isDescriptor concept to restrict the allowed types for the template
- Classes encapsulation to comfortably access each object when needed

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